Using the Law of Conservation of Energy to solve problems

Student worksheet

Use $g = 9.8\ \text{m s}^{-2}$ throughout unless stated otherwise.

Working standard: whenever you use an equation, write the equation symbolically first, then rearrange if necessary, then substitute values, then state the answer with units.

Core principle

Energy cannot be created or destroyed. For a suitably chosen system, total energy before an event equals total energy after the event.

Total energy before = Total energy after

Main AP quantities: K, ΔU_g, U_G, U_s, ΔU_{int}.

Task 1 · Identify the main energy transfer

1. A small, dense ball falls through the air.

Energy type 1: _____ → Energy type 2: _____

2. A car with its engine switched off coasts uphill and slows down.

Energy type 1: _____ → Energy type 2: _____

3. The school lift descends at steady speed.

Energy type 1: _____ → Energy type 2: _____

4. A compressed spring launches a cart on a horizontal track.

Energy type 1: _____ → Energy type 2: _____

5. A bicycle brakes to rest on level ground.

Energy type 1: _____ → Energy type 2: _____

6. A bouncy ball strikes the floor and compresses before rebounding.

Energy type 1: _____ → Energy type 2: _____

GREENSICK bank: Gravitational · Radiation · Elastic · Electrostatic · Nuclear · Sound · Internal · Chemical · Kinetic

Task 2 · Simple conservation

A. A 100 W lightbulb converts 100 J of electrical energy each second. If 75 J becomes internal energy each second, what other energy change can be deduced?

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B. A ball loses 250 J of gravitational potential energy and gains 200 J of kinetic energy. Suggest what else has occurred.

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C. A stretched elastic band is released vertically upward. At the highest point it has gained 25 J of gravitational potential energy. Neglect losses. What can you deduce about its initial elastic potential energy?

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Task 3 · Gravitational and kinetic energy

A 28 g ball falls from rest from a table 78 cm high.

(a) Determine Δy, including sign.

(b) Write $\Delta U_g = mg\Delta y$ first, then calculate ΔU_g.

(c) Explain the sign.

(d) Write a suitable kinematics equation first, then determine speed.

(e) Write $K = \frac{1}{2}mv^2$ first, then determine K.

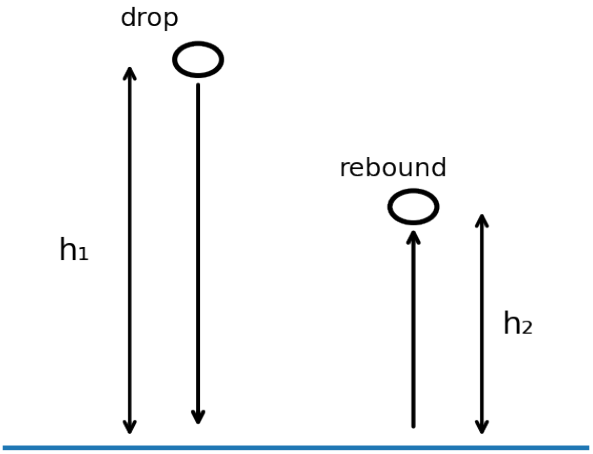
(f) Compare the kinetic-energy change with ΔU_g.

Task 4 · Before-and-after accounting

$$\mathbf{K_i + U_{g,i} + U_{s,i} + U_{int,i} = K_f + U_{g,f} + U_{s,f} + U_{int,f}}$$

For a ball dropped from rest with negligible drag, write the reduced equation:

Task 5 · Practical: bouncing ball



Equipment: bouncy ball, mass balance, metre stick, stopwatch.

Measure m, h_1, and at least five values of h_2. Average h_2.

m = _____ kg h_1 = _____ m

h_2: 1 _____ 2 _____ 3 _____ 4 _____ 5 _____ mean _____

Write $U_{g,i} = mgh_1$ first, then calculate.

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Write $U_{g,rebound} = mgh_2$ first, then calculate.

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Write $\Delta E_{mech,not\ recovered} \approx mg(h_1 - h_2)$ first, then calculate.

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Discuss energy transfers and uncertainty.

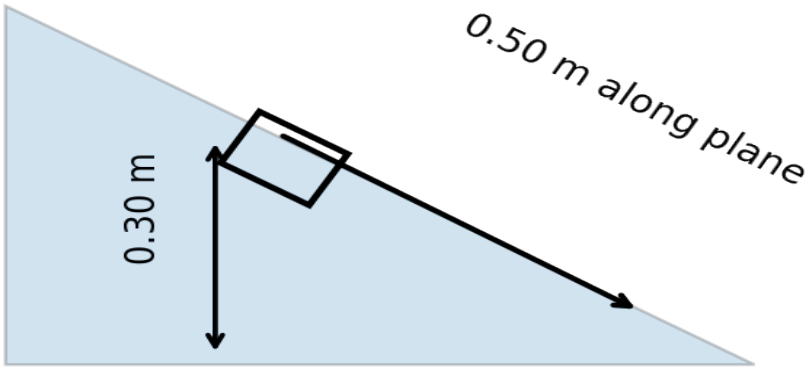
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Task 6A · Compare Force & Motion with Energy



A 1.2 kg block starts from rest and moves 0.50 m down a frictionless incline while dropping 0.30 m vertically.

(a) Determine θ.

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(b) Write $F_{g,\parallel} = F_g \sin\theta$ first, then calculate.

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(c) Write $\Sigma F_{\parallel} = F_{net} = ma$ first, then find a.

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(d) Write the kinematics equation first, then find v.

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(e) Write $\Delta U_g = mg\Delta y$ first, then calculate.

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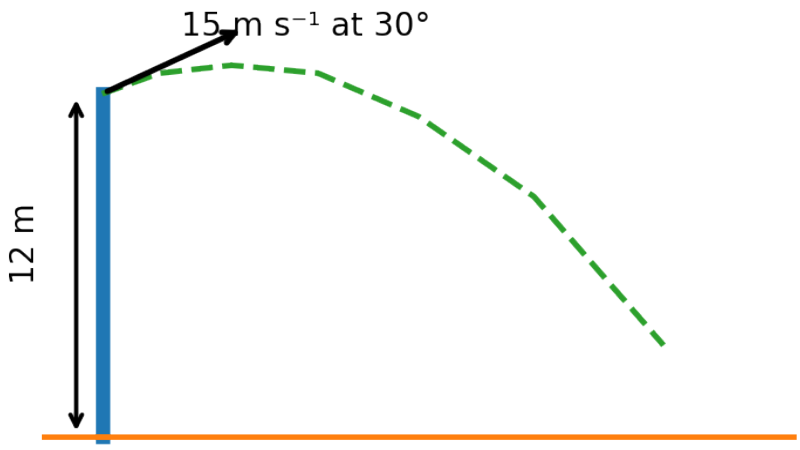
(f) Write $K = \frac{1}{2}mv^2$ first, then calculate.

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Task 6B · Projectile



A 0.25 kg ball is launched at 15 m s^{-1}, 30° above horizontal, from a building 12 m high. Air resistance is negligible.

(a) Find $v_{x,0}$.

(b) Find $v_{y,0}$.

(c) Use a kinematics equation to find rise above roof.

(d) Find K_i.

(e–g) At the top, find v_x, speed and K.

(h–i) Find ΔK and ΔU_g from launch to top.

(j–l) Find ΔU_g roof→ground, then impact K and speed.

Task 7 · General gravitational potential energy

Near Earth use $\Delta U_g = mg\Delta y$. Over large distances use $U_G = -Gm_1m_2/r$.

Use $G = 6.67 \times 10^{-11}$ N m² kg^{-2}, $M_E = 5.98 \times 10^{24}$ kg, $R_E = 6380$ km, $m = 65$ kg.

(a) Find U_G at R_E.

(b) Find U_G at $2R_E$.

(c) Find ΔU_G.

(d) Find escape energy.

(e) Apply $\Delta U_g = mg\Delta y$ over 6380 km and explain why it is invalid.

Task 8 · Dropping a ball with drag

A 520 g ball falls from rest through 14 m.

(a) Write $|\Delta U_g| = mgh$ first, then calculate.

(b) If all becomes K, find K.

(c) Write $K = \frac{1}{2}mv^2$ first, then find v.

(d) If half becomes ΔU_{int}, find v.

(e) Write the kinematics equation first, then find a.

(f) Take down as positive. Start with $\Sigma F = F_{net} = ma$, then determine F_{drag}.

Task 9 · Generalise by deriving

(a) Mass M starts from rest and falls height H. Begin with $K_i + U_{g,i} = K_f + U_{g,f}$ and derive v.

(b) Repeat for initial downward speed v_0.

Task 10 · Projectiles

1. A 2.5 kg object is projected horizontally at 5.0 m s^{-1} from a 12 m building.

(a) Find K_i.

(b) Find ΔU_g and hence ΔK.

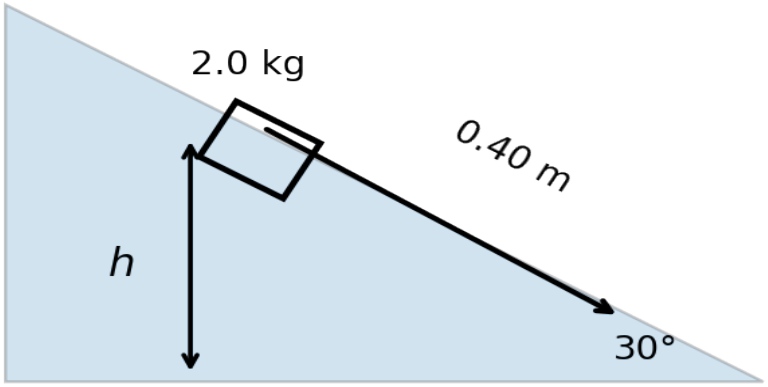
(c) Find K_f and impact speed.

2. Generalise for M, v_0, H, beginning with conservation of energy.

3. A 0.500 kg object is projected at 12 m s^{-1}, 60° above horizontal.

(a–g) Find initial K, v_x, top-point K, ΔK, ΔU_g, and maximum height. State each equation before substitution.

Task 11 · Inclined planes



A 2.0 kg block starts from rest on a frictionless 30° plane and moves 0.40 m down it.

A. Force method: write $F_{g,\parallel} = F_g \sin\theta$; then $\Sigma F_{\parallel} = F_{net} = ma$; then a kinematics equation.

B. Energy method: write $h = d \sin\theta$; then $K_i + U_{g,i} = K_f + U_{g,f}$; then $K = \frac{1}{2}mv^2$.

C. Generalise for M, d, θ.

Task 12 · Challenge algebra: rough incline

A block of mass m starts from rest and slides down a rough plane inclined at θ. Vertical drop h; coefficient of kinetic friction μ_k. Derive expressions using only m, θ, h, μ_k, g.

(a) d.

(b) F_n; begin with $\Sigma F_\perp = 0$.

(c) F_f; begin with $F_f = \mu_k F_n$.

(d) F_{net} down plane; begin with $\Sigma F_\parallel = F_{net}$.

(e) a; begin with $F_{net} = ma$.

(f) v; begin with $v^2 = v_0^2 + 2a\Delta x$.

(g) K_f; begin with $K = \frac{1}{2}mv^2$.

(h) ΔU_g; begin with $\Delta U_g = mg\Delta y$.

(i) ΔU_{int}; begin with conservation of energy.

Questions I still have